When Operator Accuracy Does Not Transfer to the Image: A Learned Boundary Transport Operator for Faceted Gemstones

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Abstract—Neural appearance models compress multiple scattering into a small learned function and work well for fibrous media such as cloth, where the scattering structure lies far below the scale of a pixel. We ask whether this approach transfers to the opposite case: a faceted diamond, whose light transport is specular and discrete, and whose fine structure is its appearance. We show that the radiance distribution map (RDM) of the source method conditions on the wrong variable for this geometry, and generalise it to a learned boundary operator: a network that takes the entry position, direction, surface normal and wavelength of a light path and replaces its whole journey inside the stone with one prediction of the exit facet, point and direction. The operator learns, outperforming every non-learned baseline including a nearest-neighbour oracle over a million paths, yet it does not reproduce the image. Across three cuts trained on identical data, the more accurate deterministic operator renders sharper, more diamond-like images that agree with an exact reference less than a blurred stochastic one (correlation 0.038–0.168 against 0.136–0.293). Data volume, capacity, training distribution and input encoding are excluded as causes; the failure splits into exit-facet selection and within-facet placement, and fixing either alone makes the image worse. We further show that a whole-image score can be produced entirely by light the network never touches, and argue that the limit lies in representing a discontinuous map with a smooth function.

Index Terms—neural rendering, light transport, appearance modelling, gemstones, mixture density networks, evaluation

I. Introduction

A diamond's appearance is made almost entirely inside it. Light that enters the crown is refracted, reflected internally several times and refracted out again, and each facet acts as a mirror onto a different part of the surroundings. Small changes in viewpoint move these reflections across facet edges, which is what produces brilliance, fire and scintillation. Rendering this by path tracing is exact but repeats the whole interior journey for every sample.

Neural appearance models offer a way to amortise such journeys. Soh and Montazeri [1] compress the multiple scattering inside a yarn into small networks trained from a radiance distribution map (RDM): a table, gathered by tracing the true transport, of how light leaves for each incoming direction. For cloth this is well founded: hundreds of fibres sit inside one pixel, so the aggregated appearance really is smooth.

A diamond inverts every one of those conditions. It has a few dozen facets, each resolved across thousands of pixels; one entry state produces a few exact, coherent beams rather than a lobe; and its appearance consists precisely of the angular detail that aggregation discards. It is therefore an adversarial test of whether the aggregate approach transfers. Our research question is: do aggregate neural appearance models transfer from fibrous media to resolved specular media, where the fine structure is the appearance?

Our answer is no, and we characterise why. The contributions are:

1) A measured diagnosis that the RDM conditions on direction where a faceted stone's exit depends on position (Sec. III).
2) A learned boundary operator that generalises the RDM to continuous conditioning on position, direction, normal and wavelength, with stochastic and deterministic exit decoders (Sec. IV).
3) An evaluation apparatus built to make a negative result credible: measured noise floors, an exact split of each image into light that did and did not pass through the network, and operator-level metrics separated from image metrics (Sec. V).
4) The central finding, controlled across three cuts, that operator accuracy does not transfer to the image, with an elimination of data, capacity, training distribution and input encoding as causes (Sec. VI).

II. Related Work

Aggregate neural appearance. Soh and Montazeri [1] fit small networks to the multiple-scattering component of yarn appearance, conditioning on incoming and outgoing

directions and treating the direct reflection analytically. Our starting point was an adaptation of their method to diamond.

Learned transport through a region. Vicini et al. [2] learn where light exits a translucent object given where it enters, using a conditional variational autoencoder and a local polynomial shape descriptor, and explicitly assume a diffuse, separable outgoing direction. Kallweit et al. [3] replace multiple scattering in clouds with a network. Both address diffusive media, where the exit distribution is smooth; we study the specular regime those works exclude, in which the exit direction must be learned and is a discontinuous function of the entry state.

Recovering variation lost to averaging. LEAN mapping [4] stores first and second moments so that filtered surfaces keep their roughness, and stochastic glint models [5] anchor sparkle to surface position. We adapt the moment idea to the RDM before moving away from it.

Networks and sampling. Our stochastic decoder is a mixture density network [6]; our deterministic decoders follow behavioural cloning [7]. Multilayer perceptrons are biased towards smooth functions [8], and Fourier features [9] and multiresolution hash encodings [10] are standard remedies. Learned densities can serve as importance-sampling proposals [11] when mixed defensively [12], [13]. Explicit gemstone rendering [14] and physically based rendering [15] provide our reference; all rendering uses Mitsuba 3 [16] with hero-wavelength spectral sampling [17].

III. From Radiance Maps to a Boundary Operator

The RDM. The source representation bins how light leaves the stone by direction for each incoming direction, compresses the table into small networks, and applies the result as a material on the stone's surface. Rendered on a round brilliant it produced flat, blocky shading.

Averaging erased the detail. Each RDM bin stores the mean brightness of the paths that land in it. Measured over the gathered paths, the standard deviation exceeded the mean in 99.8% of well-populated bins, with a median ratio of 2.53: each stored value summarises a distribution several times wider than itself.

A LEAN-inspired sparkle. Following LEAN mapping [4], we stored the second moment of each bin and turned shading points on with probability $p = 1/(1 + r^2)$ at gain $1/p$, where r is the bin's relative spread, keyed to surface position by a hash so the pattern stays fixed as samples accumulate. This preserves mean brightness exactly and restored the amount of variation, but not its placement: the bright points are statistically right and spatially arbitrary, and the result read as speckle.

The missing variable is position. Asked to predict the exit facet on held-out paths, a table conditioned on entry direction, as the RDM is, scores 0.068 top-1 against a trivial baseline of 0.047; enlarging it from 32 to 2,048 cells raises it only to 0.098. A table conditioned on the entry facet, a proxy for position, reaches 0.192 (Table I). Resolution is not the limit; the conditioning variable is. In yarn light enters and leaves within a sub-pixel cross-section, so direction suffices; in a brilliant cut, where light enters decides which facet it leaves through.

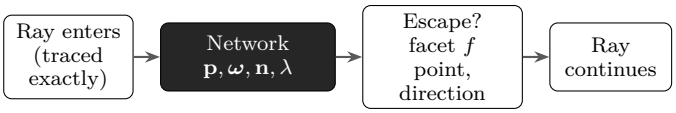


Fig. 1. The learned boundary operator. One network query replaces every internal reflection and refraction; everything outside the stone remains ordinary path tracing, identical in both modes.

IV. Method

A. Scene and reference

One studio scene is used throughout: a perspective camera with a 30° field of view, a grey diffuse ground plane, textured area lights and two small intense sources. Structured lighting is essential: under a constant environment every exit direction returns the same radiance and any transparent object renders as featureless grey. The analytic reference is a spectral path tracer in which the stone is a smooth, dispersive dielectric: at each interaction, reflection or refraction is chosen by the Fresnel coefficient, directions follow Snell's law with a Sellmeier index for diamond, and paths are traced until escape or a depth of 128. The estimator samples the BSDF only. We use three procedurally generated cuts, all validated as closed, consistently oriented meshes: a GIA-proportioned round brilliant (64 triangles), a pear brilliant (62) and a step cut (112).

B. The learned boundary operator

When a camera path refracts into the stone, its entry position $\mathbf{p}$, direction $\boldsymbol{\omega}$, surface normal $\mathbf{n}$ and wavelength λ are known. The operator maps these ten numbers, $\mathbf{x}$, to the outcome of the entire interior journey (Fig. 1):

$$p(\text{exit} \mid \mathbf{x}) = p(e \mid \mathbf{x})\, p(f \mid \mathbf{x})\, p(\mathbf{y} \mid \mathbf{x}, f), \tag{1}$$

where e is a Bernoulli escape event, f a categorical over the mesh's triangles, and $\mathbf{y} \in \mathbb{R}^4$ the exit coordinates on facet f. Where the RDM answers light arriving from this direction leaves in these directions, the operator answers light entering here, this way, at this wavelength, leaves at this point on this facet, in this direction.

Exit coordinates. The exit point is written as two barycentric logits on facet f and the exit direction as two slopes in the facet's tangent frame $(\mathbf{t}, \mathbf{b}, \mathbf{n}_f)$:

$$\mathbf{d} = \frac{\mathbf{n}_f + y_3\mathbf{t} + y_4\mathbf{b}}{\|\mathbf{n}_f + y_3\mathbf{t} + y_4\mathbf{b}\|}. \tag{2}$$

A softmax over the logits gives positive barycentric weights summing to one, so every prediction decodes to a point on the facet and an outward direction.

Architecture. A shared encoder (two layers of width 64, SiLU [18]) feeds an escape head and a facet head; a learned 16-dimensional facet embedding conditions the exit decoder. We compare three decoders, sharing everything else:

- Stochastic: a mixture density network [6] with four Gaussian components over $\mathbf{y}$, trained by negative log-likelihood and sampled at render time (17,637 parameters on the round cut).
- Clone: a two-layer regressor predicting one exit, trained with a smooth-L1 loss (behavioural cloning [7]; 15,557 parameters). It isolates sampled versus committed placement as the single variable.
- Deep decoder: the clone with a four-layer regressor reading its own encoder, the best configuration of a decoder sweep (28,741 parameters).

In all three the exit facet is sampled from the categorical; they differ only in how the exit is placed on it.

Data and training. Training paths are produced by the analytic tracer: rays are fired at the stone, refracted in and traced to escape, and each record stores the entry state, exit facet, point, direction and an escape status. Entry states are drawn to match the camera's query distribution over all azimuths (Sec. VI-D revisits this). Each pool holds 1,048,576 paths from 65,536 entry states, with the split grouped by entry state so that repeated outcomes of one entry never straddle it. Models are trained with Adam [19] (learning rate 10^{-3}, batch 256), stochastic decoders for 80 epochs and deep decoders for 30, keeping the best validation checkpoint.

Rendering. When a camera path hits the stone, Fresnel reflection or transmission is sampled exactly. On transmission the entry state is batched with 511 others and sent to the operator; each returned exit spawns a new ray into the scene. An exit that would immediately re-enter the stone is discarded rather than corrected.

V. Evaluation Methodology

A negative result is only as credible as the apparatus that produced it.

Paired renders and noise floors. Neural and analytic images are rendered at identical settings (512×512, box filter) as independent seeds of 32 samples per pixel. Averaging independent seeds is the same estimator as one render at the combined count. Splitting the seeds into two halves gives two estimates of the same image; their agreement is the noise floor, the best any method could score at that sample count, and the neural–analytic comparison is made on matching halves.

Image metrics. All metrics are computed over pixels whose primary ray hits the stone. Correlation is the Pearson correlation of linear luminance between the neural and reference images, measuring whether light is in the right place irrespective of overall brightness; we also report highlight overlap (the share of the brightest 1% of pixels in common), relative energy error, and contrast (standard deviation over mean).

TABLE I
Exit-facet prediction on held-out paths (round cut).

Predictor	Top-1
Always the most common facet	0.047
RDM-style table, entry direction (32 cells)	0.068
Table, entry facet (64 cells)	0.192
Nearest-neighbour oracle, top-1 / majority of 8	0.287 / 0.343
Operator, stochastic decoder	0.380
Operator, clone decoder	0.410

Network-only split. Every camera path is tagged by whether it passed through the operator, splitting each frame exactly into a network component and an untouched component. The untouched component is identical physics in both modes and serves as a control: it agrees between modes at 0.958–0.961 in every run. The network component carries 85.5% of the stone's energy.

Operator-level metrics. Measured on held-out paths without rendering: exit-facet top-1 accuracy (branch selection), and the angle between the predicted and true exit direction when the decoder is given the correct facet (within-facet placement). A nearest-neighbour oracle over the training paths tests whether the exit is predictable from the inputs at all, independently of any architecture.

VI. Results

A. The operator learns

The operator exceeds every non-learned baseline (Table I), including a nearest-neighbour oracle over 1,046,359 training paths by 42%, and predicts escape almost perfectly (0.9984 predicted against 0.9975 measured). A conditioning ablation, zeroing one input group at a time with the architecture fixed, confirms the diagnosis of Sec. III: removing position drops top-1 from 0.404 to 0.273, while an operator given only the entry direction, the RDM's conditioning, falls to 0.117. Removing wavelength costs 0.0004.

B. Sharper is not more accurate

Fig. 2 and Table II give the central result. On each cut the operators were trained on the identical pool, so the decoder is the only variable. The deterministic decoders are the better operators: they choose the exit facet more often (validation top-1 0.392 against 0.371 on the round cut, 0.337 against 0.309 on the pear, 0.328 against 0.305 on the step cut) and place exits closer to the truth. Their images look more like a diamond, with contrast approaching or, on the pear, exceeding the reference's. Yet on every cut they agree with the reference less, by a factor of 1.7 to 4. Fig. 3 shows the same pattern on the pear and step cut.

The stochastic decoder is the opposite case. It gets the stone's outline and total energy right (within 1% on the

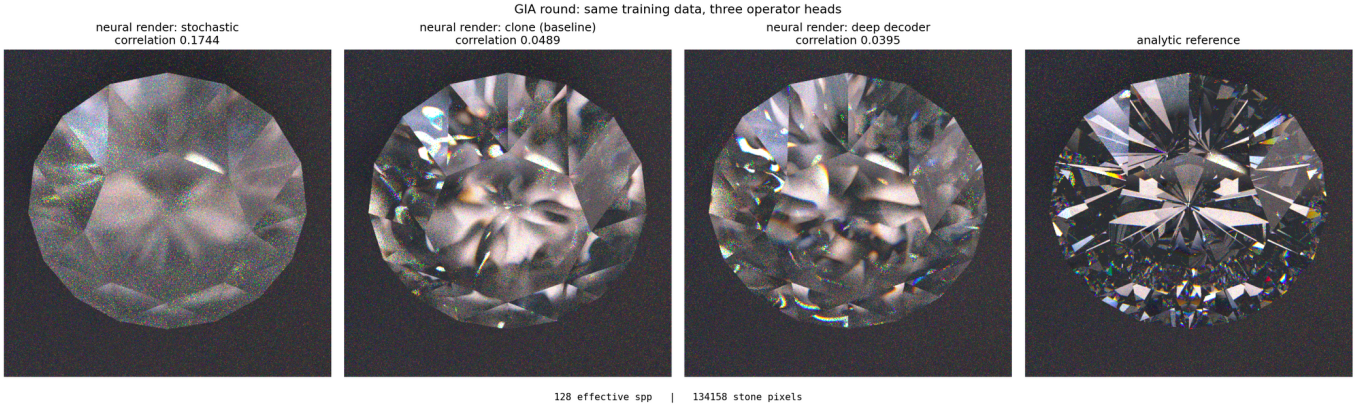


Fig. 2. Round brilliant rendered with three operators trained on the same paths, beside the exact reference (128 samples per pixel). The deterministic decoders produce sharp, high-contrast highlights in a melted pattern; the stochastic decoder is soft and flat but agrees best with the reference. Panel labels give full-frame correlation.

TABLE II
Same training data per cut; only the exit decoder differs. Split-half correlation with the analytic reference (noise floor in brackets), relative energy error and contrast (reference in brackets). 128 samples per pixel.

Cut	Decoder	Corr.	Energy	Contrast
Round (64)	stochastic	0.152	−0.8%	0.93
	clone	0.048	+26%	2.35
	deep decoder	0.038	+14%	2.74
	floor / ref.	(0.957)		(3.78)
Pear (62)	stochastic	0.136	+17%	0.89
	deep decoder	0.054	+48%	2.31
	floor / ref.	(0.899)		(1.48)
Step (112)	stochastic	0.293	+14%	0.84
	deep decoder	0.168	+15%	1.77
	floor / ref.	(0.942)		(2.37)

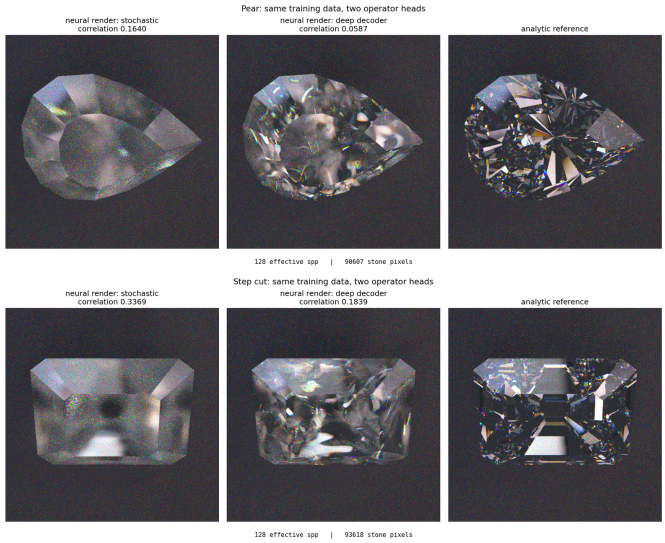


Fig. 3. Pear (top) and step cut (bottom): stochastic and deep-decoder operators beside the reference. The broad layout survives on both; the sharp virtual facets do not.

round cut), but its exits are spread so widely that each pixel averages bright and dark parts of the lighting, and the stone renders with under a quarter of the reference's contrast, like frosted glass. In a doubling test (128 to 256 samples per pixel), noise fell at the expected rate while the image's structure was unchanged: the errors are bias, not noise.

C. Why: what was ruled out

Table III summarises the elimination chain. Two results stand out. The trained network already beats local interpolation, yet a substantially deeper decoder closes only 18.6% of the gap to a nearest-neighbour lookup in within-facet placement, so the limit is not how much the decoder can compute. And Fourier input features, the standard remedy for spectral bias, made the operator worse at every scale tested; giving the stochastic operator the same encoding left its image unchanged (correlation 0.168 against 0.174).

The nearest-neighbour analysis shows why interpolation fails. Eight near-identical entry states span a median of two distinct exit facets, and all eight agree for only 18.6% of queries: entry states 4.7° apart routinely leave through different facets. The map from entry to exit is piecewise smooth with dense discontinuities; within a branch it is smooth (the oracle places exits within 8.9° when it finds the right facet), but the branch boundaries are finer than any achievable sampling.

Two failures, not one. To separate them we let the exact tracer supply the true exit facet for every path, leaving the network only the within-facet placement (Fig. 4). The stone's structure returns: contrast is 3.73 against the reference's 3.74 and peak-to-mean 115 against 121. Correlation nonetheless falls to 0.041, because each shard is still misplaced by roughly 20°. Facet choice determines whether the image looks like a diamond; within-facet placement determines whether it is the right one; the

TABLE III
Elimination chain (round cut unless stated).

Hypothesis	Evidence
Data	Top-1 rises from 0.048 to 0.332 between 256 and 16,384 entry states, then only to 0.347 at 65,536.
Capacity	Width 256 gains 0.33 nats, then overfits; a four-layer decoder reduces placement error from 24.5° to 21.6°; a parameterless lookup reaches 8.9°.
Training distribution	Camera-matched data fixed a +15.8% energy bias but not placement.
Density sharpness	Decoding at the component mean cuts spread from 26.0° to 8.1°; correlation falls 0.174 → 0.153.
Input encoding	Fourier features [9] at three scales: placement 25.8°–26.7°, worse than the 21.6° control.
Geometry	The failure recurs on pear and step cuts (Table II).

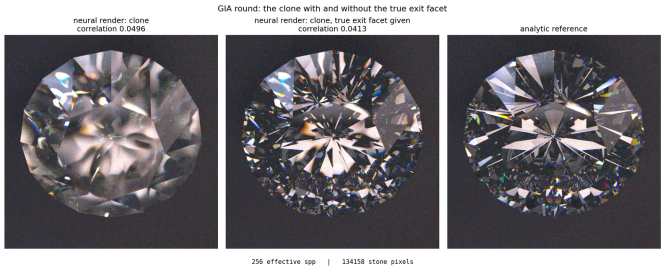


Fig. 4. Clone decoder without (left) and with (centre) the true exit facet supplied by the exact tracer, beside the reference (256 samples per pixel). The correct facets restore the star pattern and contrast, but not pixel agreement.

operator fails at both.

D. Training where light enters

Camera-matched training entries all land on the crown: measured with the real sampler, 0% enter through the pavilion (Fig. 5, top). Tumbling the pear through 60 frames exposed the consequence: when the pavilion faced the camera, the operator let only 46% of queries escape and the underside went dark. Adding entries drawn uniformly over directions, each offset across a disc the size of the stone, places 53% of entries on the pavilion and restores escape to 100% at every angle (Fig. 5, bottom) and energy to within 28%. Placement, however, was not fixed: correlation at a pavilion-facing pose remained 0.018 against a floor of 0.951. Coverage is necessary but not sufficient.

E. Transfer, and a warning about whole-image scores

The operator's facet categorical is tied to its training mesh. Re-pointing the pear operator at an unseen round brilliant with re-angled facets (crown 33.0°, pavilion 41.2°, table 59%), matching facets by normal, gave a correlation of 0.37 for the stochastic and 0.35 for the deep decoder, the highest of any render. The network-only split shows that

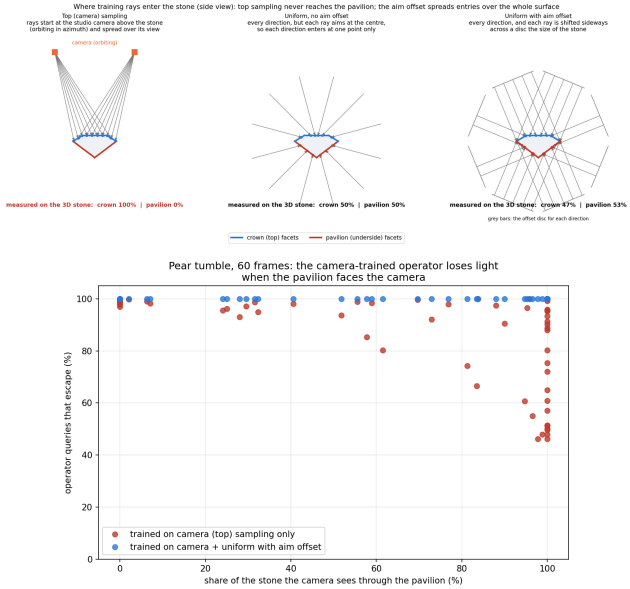


Fig. 5. Top: where training rays enter the stone under camera sampling, uniform sampling, and uniform sampling with an aim offset. Bottom: escape rate against the share of the stone seen through the pavilion, for every frame of a 60-frame tumble, before and after adding uniform entries.

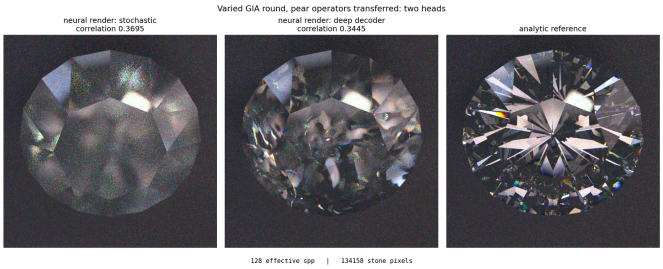


Fig. 6. A pear-trained operator on an unseen round brilliant. The bright oval on the table is an exact surface reflection shared by both images; it alone produces the apparent agreement.

this score is an artefact (Fig. 6). The flatter crown reflects one lamp directly into the camera; 90% of that highlight's light never passed through the network, and removing an 81×81 patch around it drops correlation to -0.001. The network's own contribution correlates with the reference at 0.004. On the default round cut, the same patch removal leaves the native operator's correlation unchanged (0.205), so its agreement is genuinely distributed. Whole-image metrics for learned transport should therefore be reported alongside the network-only component.

F. Dispersion and cost

The operator ignores its wavelength input, which a paired-wavelength experiment explains. Tracing each path at four wavelengths that replay the same reflect/transmit decisions, all colours keep the same route on 94% of single-interaction paths but on only 16% of paths with ten or more interactions. Where routes coincide, colours leave

a median 0.37° apart, more than fifty times below the operator's placement error, so the wavelength signal lies beneath the noise. A separate head that predicts whether each colour stays on the hero wavelength's route and, if so, its exit shift explains 84.6% of the colour spread on held-out pairs; rendered, it doubles coherent colour (0.12 to 0.23, against 1.01 for the reference) and cuts colour noise by two thirds, but true fire still rides on misplaced exits.

The operator does not save time. A query costs 5–9 μs, but in our CPU renderer the surrounding per-ray bookkeeping makes neural renders 1.5–2 times slower than exact tracing, which needs only about nine ray–triangle interactions per path. With little transport to amortise, the speed premise is weak in this domain before accuracy is considered.

VII. Discussion

Why improvements make the image worse. Correlation partly rewards diffuse overlap. A blurred operator spreads each facet's light over a wide cone, so some of it always lands where the true light does; a sharp operator concentrates the same energy into narrow beams, which score well only if aimed correctly. Accuracy helps the image only once facet choice and within-facet placement are both good enough for sharp light to land in the right place. Below that threshold, every step towards sharpness is a step away from agreement, which is why the deterministic decoders, the deterministic decode and the oracle facet all moved the image the wrong way.

Why the stochastic operator is flat. Each sample draws one exit; under a broad exit distribution, samples from one facet fan across many parts of the lighting rig, and their average approaches the rig's mean radiance, the single condition under which a transparent object looks featureless. The learned operator thereby converts a structured environment into an effectively constant one, accounting for the milkiness and absent scintillation from one cause.

Where the limit lies. Data, capacity, training distribution and a global input encoding are excluded, and the network already exceeds local interpolation. The remaining explanation is representational: a diamond's exit map jumps wherever an internal path crosses a facet edge, and a small smooth network blends across those jumps instead of reproducing them. Local encodings that let sharp changes stay local [10], and shape descriptors that let one operator see facet geometry rather than a fixed facet index [2], are the directions this evidence points to.

Was the pivot from the RDM valid? The RDM was diagnosed, generalised and then measured. Used as an importance-sampling proposal it increases variance by a factor of 1.3 at equal samples; used as a prior on the exit facet its best gain is +0.003. It is a marginal of the conditioning the operator already receives, so it cannot add information the operator lacks. It remains the right model where its assumption holds: media aggregated below the pixel.

VIII. Limitations

Comparisons use one studio rig and mostly one view, at 512×512. Agreement is measured over stone pixels only, so caustics on the ground are not evaluated. Each configuration was trained once, without training-seed error bars, and the deep decoders were still improving at 30 epochs, so their operator-level advantage is likely understated. The meshes are simplified (62 to 112 triangles), and the facet head is sized to its mesh, which limits transfer. The reference's convergence with sample count has not been demonstrated separately from its split-half agreement. Correlation is our primary image metric; perceptual metrics such as SSIM or ꟻLIP would complement it.

IX. Conclusion

Aggregate neural appearance models do not transfer to resolved specular media, and the reason can be measured rather than asserted. The RDM conditions on direction where a faceted stone depends on position; generalising it to a position-conditioned boundary operator yields a network that learns more than any lookup, yet cannot reproduce the image. Across three cuts with the decoder as the only variable, the more accurate operator renders a more convincing and less faithful diamond. For media whose fine structure is the appearance, the question is not whether a network can learn the transport approximately, but whether it can learn it precisely enough for that precision to be visible.

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